

SMART INDIA HACKATHON · PS 26146 · NTRO

BEANS

Bitcoin Encryption, Analysis & Network Security

Technical report: implementation, features and measured accuracy

0.967

fused-risk PR-AUC, mean
of 6 synthetic datasets

98.4 %

of criminal entities alerted

0.799

illicit F1 on real Bitcoin
data (Elliptic)

0.98 M

rows loaded + scored in
286 s

Release v1.0 (commit 239d3ea) · generated 2026-09-26 · runs fully offline

All figures in this report are generated from result files in the repository (`scripts/build_report.py`).

1. Summary

BEANS ingests bulk Bitcoin transaction and network-relay observations (CSV, JSON or XML), enriches every relaying IP offline with country and ASN type, and links IPs, transactions and wallets in one graph. Five engines score it (entity clustering, anomaly detection, transaction-shape classification, risk propagation from known illicit seed wallets, and a graph neural network); a calibrated fusion model turns them into a ranked, explained alert list. Every alert then gets a **recommended next step** from fixed, citable rules: draft a freeze request or a Section 94 BNSS notice, prepare an FIU-IND referral, watch the funds, or review a likely false positive. Legal drafts need a second person's approval; evidence is sealed with SHA-256 and an offline RFC 3161 timestamp; critical alerts can be pushed to a SIEM.

0.935

recall of illicit wallets ($P \geq 0.5$), before 0.844

0.960

precision ($P \geq 0.5$), before 0.946

1.000

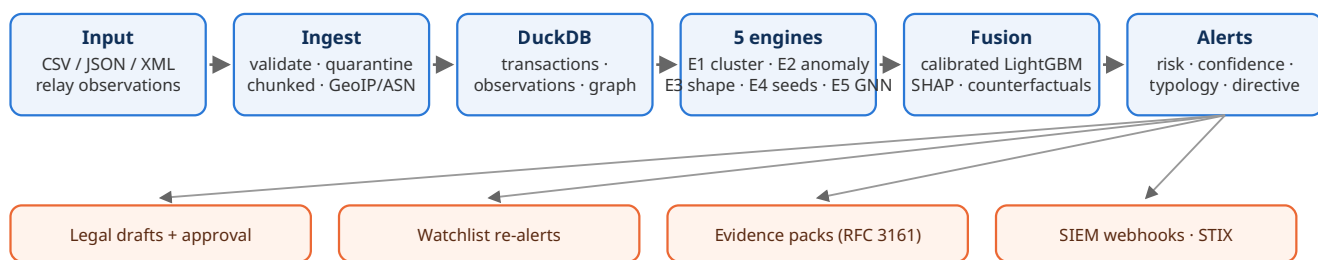
clusters never mix two criminals (all 6 datasets)

0.0041

calibration error (ECE)

How to read the numbers. Synthetic results are means over six independently generated datasets, out-of-fold and grouped by criminal entity (a model is never tested on an entity it trained on). Our generator produced that data, so absolute synthetic numbers are optimistic; the meaningful parts are the before/after comparison, the ablations and the real-data validation on Elliptic (section 5), where BEANS matches the strongest published baseline rather than beating it.

2. System overview



Runs fully offline (GeoIP, models, fonts, TSA bundled); the optional live collector is a separate tool.

Layer	What it does
Ingest	Streams CSV / JSON / XML in 50,000-row chunks; column mapping for unfamiliar exports; invalid rows quarantined with the reason, never filled with fake values; the earliest relay of a transaction wins across files (first-spy); optional wallet-software fields (version, nLockTime, RBF).
Enrichment	Offline DB-IP country and ASN databases; ASN typed as residential / datacenter / VPN / Tor exit / bulletproof hosting.
Storage	Embedded DuckDB file; investigator state (cases, verdicts, watchlist, approvals, audit trail) survives re-scoring.
Interfaces	FastAPI REST API; React dashboard (served pre-built, no Node.js needed); CLI; Docker image that passes a test with networking disabled.

3. Engines

Engine	Method	Measured (v1.0, 6-dataset mean)
E1 entity clustering	Common-input ownership (CoinJoins excluded) · change heuristics with guards (never a later-swept deposit address, never the tiny peel payment, never spent by different wallet software) · peel-chain change · self-split rule (≥ 7 near-identical fresh parts from a non-hub owner)	homogeneity 1.000 · completeness 0.744 (was 0.580)
E2 anomaly	Isolation Forest on behaviour + network features	used as a fusion input
E3 transaction shape	LightGBM on 31 structural features: peel, CoinJoin, fan-out, fan-in, round trip	macro-F1 0.997
E4 seed propagation	Personalised PageRank (both directions), decayed haircut taint with exchanges as taint sinks, hop distances incl. direction-agnostic (reaches a seed's siblings)	86 % of hidden wallets of seeded actors reached; 9.7 % of legitimate wallets (was 16.9 %)
E5 graph neural network	SIGN-style: 15 wallet signals averaged over 1- and 2-hop neighbours along and against the money flow, service hubs cut out; fusion is the readout; CPU-only, no deep-learning framework	when added: PR-AUC 0.954 → 0.966, recall 0.905 → 0.952
Fusion	LightGBM, isotonic calibration, out-of-fold by entity; money-context features (funding shape, structuring just below round amounts, deposit-like destinations); typology model with a reference corpus of 253 extra operations, pooled per cluster	PR-AUC 0.967 · ECE 0.0041
Explanations	SHAP reasons in plain English; counterfactuals ("without Tor/VPN relaying the risk would fall from 100 to 17"), flagging single-signal vs corroborated alerts	on every alert (Figure 4)

4. Accuracy on synthetic data

Protocol. `scripts/evaluate_seeds.py --seeds 42 7 123 2024 99 555` generates six independent datasets (~4,000 transactions, ~13,500 wallets, ~30 criminal entities each, only ~30 % of entities revealed as seeds) and runs the full pipeline on each in its own database. "Before" is the code at the start of the accuracy work; "v1.0" is the tagged release. Metrics are out-of-fold and grouped by entity.

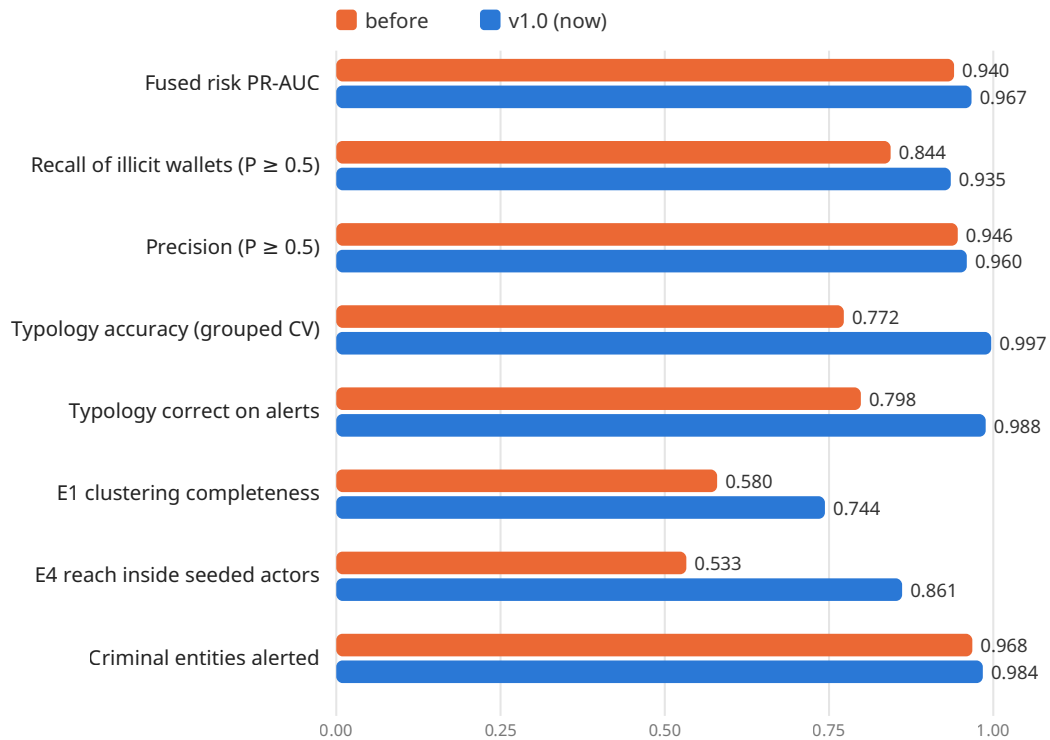


Figure 1. Before vs v1.0, mean of six datasets (values on each bar; the table below has the spread).

Metric (mean of 6 datasets)	Before	v1.0 ($\pm$ std)	
PR-AUC (fused risk)	0.940	0.967 $\pm$ 0.006	▲
ROC-AUC	0.985	0.989 $\pm$ 0.005	▲
Recall at $P \geq 0.5$	0.844	0.935 $\pm$ 0.019	▲
Precision at $P \geq 0.5$	0.946	0.960 $\pm$ 0.025	▲
Calibration error (ECE, lower is better)	0.008	0.004 $\pm$ 0.002	▲
Typology accuracy, grouped CV	0.772	0.997 $\pm$ 0.006	▲
Typology correct on alerts	0.798	0.988 $\pm$ 0.021	▲
E3 transaction-shape macro-F1	0.997	0.997 $\pm$ 0.002	
E1 completeness (illicit)	0.580	0.744 $\pm$ 0.024	▲
E1 homogeneity (illicit)	0.999	1.000 $\pm$ 0.000	▲
E4 hidden wallets reached, seeded actors	0.533	0.861 $\pm$ 0.058	▲
E4 legitimate wallets reached (lower is better)	0.169	0.097 $\pm$ 0.029	▲
Criminal entities with ≥ 1 alert	0.968	0.984 $\pm$ 0.016	▲
Alerts per criminal entity (lower is better)	9.875	5.805 $\pm$ 0.578	▲
Alerts that are illicit	0.943	0.891 $\pm$ 0.035	▼

Alert *precision* is lower than before because each criminal now takes ~5.8 alerts instead of ~9.9 (whole operations are one cluster): the same handful of false alerts weighs more in a list less than half as long, while more criminal entities are covered.

What changed, stage by stage (each a full 6-dataset run, [docs/results/stages/](#))

Stage	PR-AUC	Recall	Typology	E1 compl.	E4 seeded	Entities
start	0.940	0.844	0.772	0.580	0.533	0.968
+ money-context features	0.953	0.891	0.801	0.580	n/a	0.968
+ clustering fix, self-split, E4 fixes	0.954	0.905	0.831	0.744	0.861	0.962
+ E5 graph neural network	0.966	0.952	0.845	0.744	0.861	0.984
+ typology corpus	0.966	0.952	0.997	0.744	0.861	0.984
v1.0 (release)	0.967	0.935	0.997	0.744	0.861	0.984

- **Money-context features** (label-free): uniformity of the funding transaction's outputs, amounts just below a power of ten (structuring), hub payers, deposit-like destinations.
- **Clustering fix + self-split rule; E4 taint sinks and sibling reach**: an old change-heuristic mistake bridged whole exchanges into criminal clusters; false E4 reach fell 16.9 % → 9.7 %.
- **E5 graph neural network** and the **typology reference corpus** (253 extra operations). A label-shuffling control on the corpus (2 datasets) dropped typology to 0.48-0.61, so the gain comes from its labels; near-perfect accuracy mainly shows synthetic typologies are separable once enough operations are seen.

Tried and rejected (measured, not guessed)

- Counterparty features: PR-AUC 0.954 vs 0.953 without, dropped as no gain. Wallet-level fingerprint features: recall 0.952 → 0.922, dropped; fingerprints are kept only as a clustering guard.
- Self-split rule with ≥ 5 parts: better completeness, but merged a darknet market with its vendors; ≥ 7 keeps homogeneity at 1.00.

Demo dataset (the one the dashboard ships with)

Metric	Value
Fused PR-AUC (without network-layer features)	0.964 (0.910)
Recall / precision at $P \geq 0.5$	0.937 / 0.980
Alerts · criminal entities alerted	136 · 30 of 30
Alerts that are illicit · top 10	92 % · 100 %
Typology correct on alerts	100 %
E1 homogeneity · completeness	1.00 · 0.79
E4 reach inside seeded actors · legitimate reached	77 % · 8 %

5. Validation on real Bitcoin data (Elliptic)

Elliptic (Weber et al. 2019) has 203,769 real Bitcoin transactions (4,545 labelled illicit, 42,019 licit) and 234,355 payment edges. Its features are anonymised (no addresses, amounts or IPs), so it tests the modelling approach, not the clustering, network layer or action rules. Standard temporal split: train on time steps 1-34, test on 35-49 (1,083 illicit test transactions).

BEANS (this code)	P	R	F1	PR-AUC	Published (Weber et al. 2019)	P	R	F1
BEANS LightGBM + calibration (LF)	0.901	0.672	0.770	0.752	Logistic regression (AF)	0.404	0.593	0.481
BEANS LightGBM + calibration (AF)	0.921	0.706	0.799	0.759	Random forest (LF)	0.803	0.611	0.694
BEANS LightGBM + calibration (AF + graph)	0.909	0.713	0.799	0.760	Random forest (AF)	0.956	0.670	0.788
BEANS LightGBM + calibration (LF + GNN (E5))	0.803	0.721	0.760	0.755	Random forest (AF + GCN embeddings)	0.971	0.675	0.796
BEANS LightGBM + calibration (AF + GNN (E5))	0.853	0.718	0.780	0.756	GCN	0.812	0.512	0.628
Random forest (LF), re-run here	0.854	0.718	0.780	0.772	Skip-GCN	0.812	0.623	0.705
Random forest (AF), re-run here	0.891	0.724	0.799	0.775				

Reading. BEANS reaches illicit F1 **0.799** (precision 0.92, calibration error 0.0175), equal to the strongest published baseline and well above the published graph networks; it does not beat the random forest. E5 does not help here because Elliptic's features already are neighbourhood aggregates. Like every published model, detection collapses after time step 43, when a large dark market closed.

Seed propagation on real data. With 30 % of illicit transactions revealed as seeds, 50 % of the hidden illicit ones lie within 2 hops of a seed versus 5 % of licit ones, and adding the seeds lifts PR-AUC from 0.736 to **0.821** (the combination rule was fixed before testing).

6. Scale

	Value
Rows (10 independent files, one database)	980,803
Transactions · wallets	336,260 · 726,035
Chunked ingest (beans ingest ... --no-score)	96 s · peak 0.88 GB
Scoring, all engines (beans score)	190 s · peak 5.29 GB
End to end	286 s · 3,429 rows/s

The earlier estimate for one million rows was ~16 GB of RAM; chunked ingest and two optimised hot spots (identical results, verified to four decimals) bring it to a normal 16 GB laptop. The demo dataset scores end to end in 23 s.

7. Investigator workflow

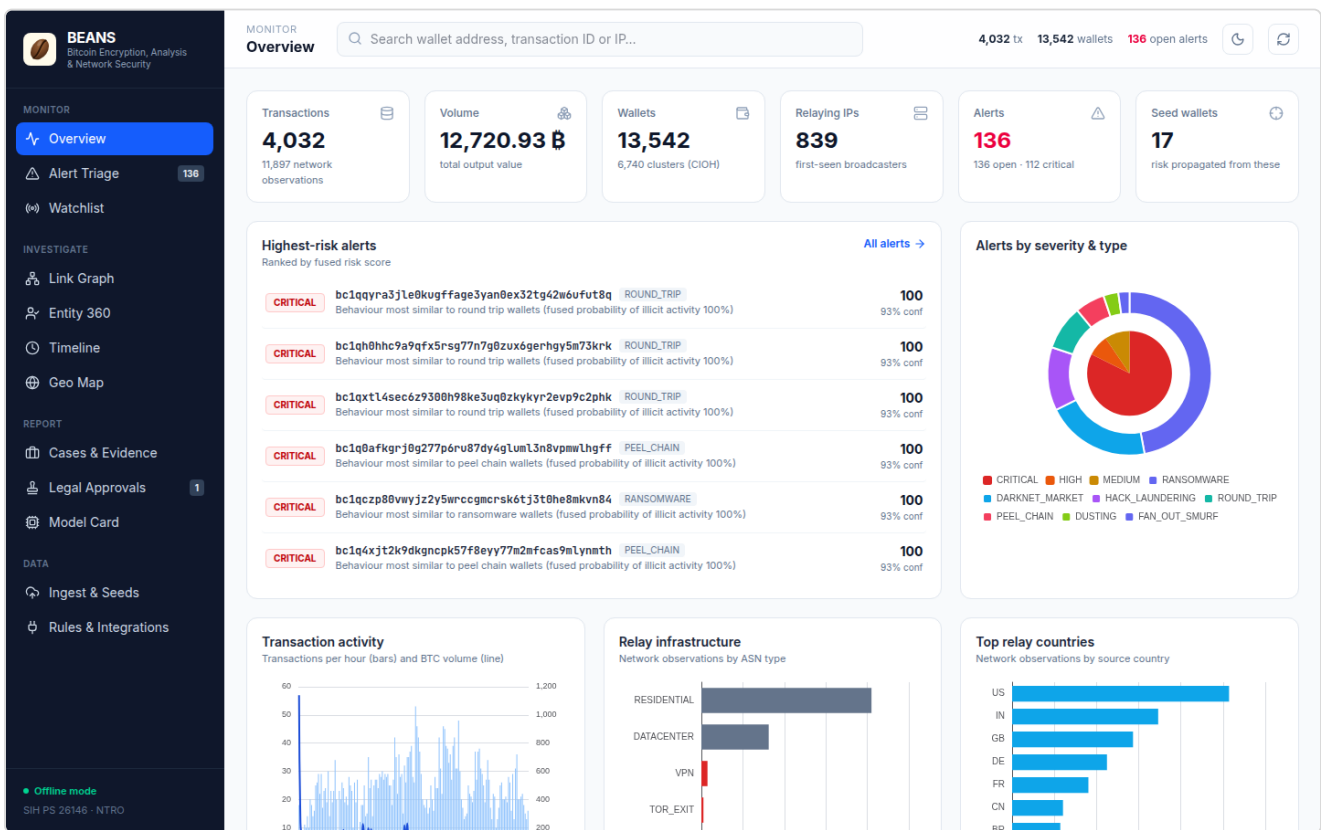


Figure 2. Overview: volume, alerts by severity and typology, relay infrastructure, countries.

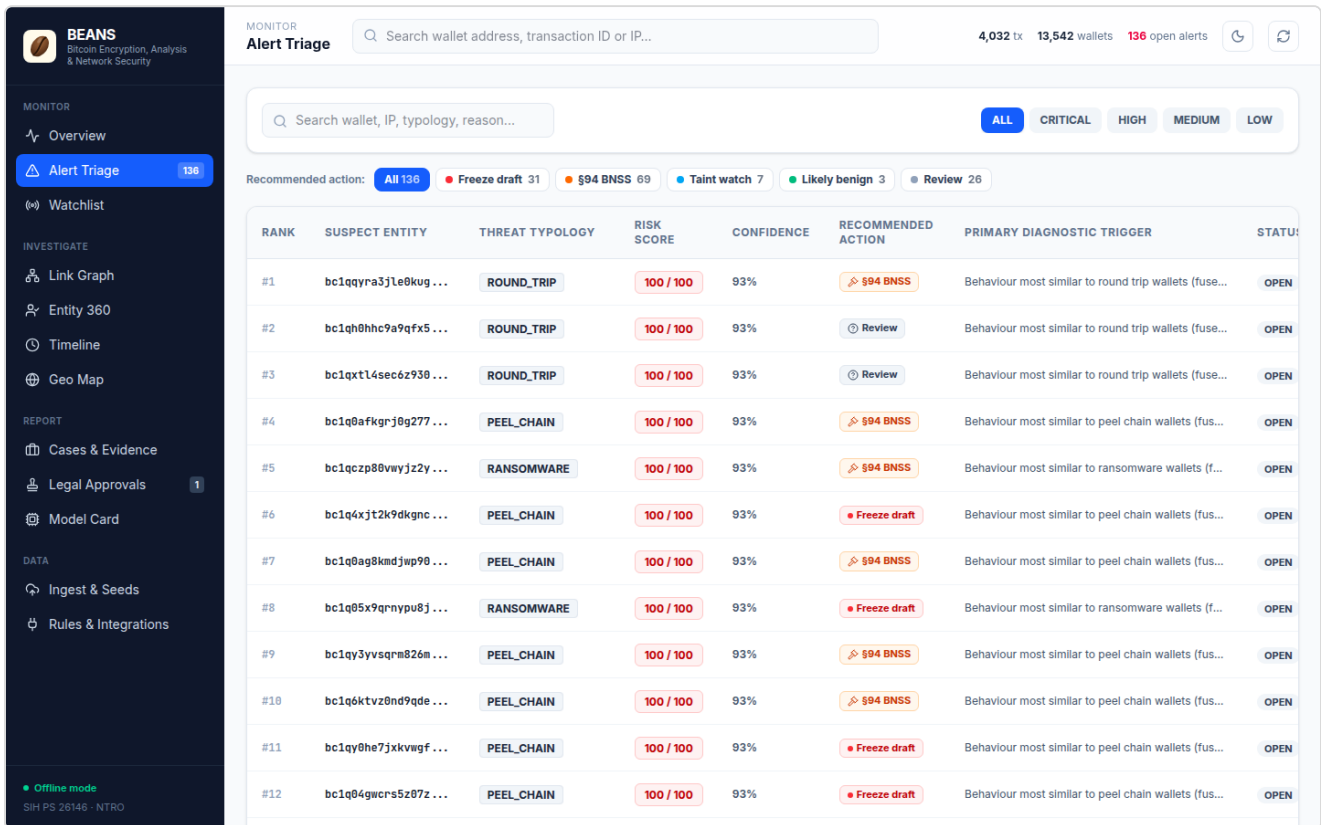


Figure 3. Alert triage: every alert carries a recommended action; filter chips by action; linked-case badges.

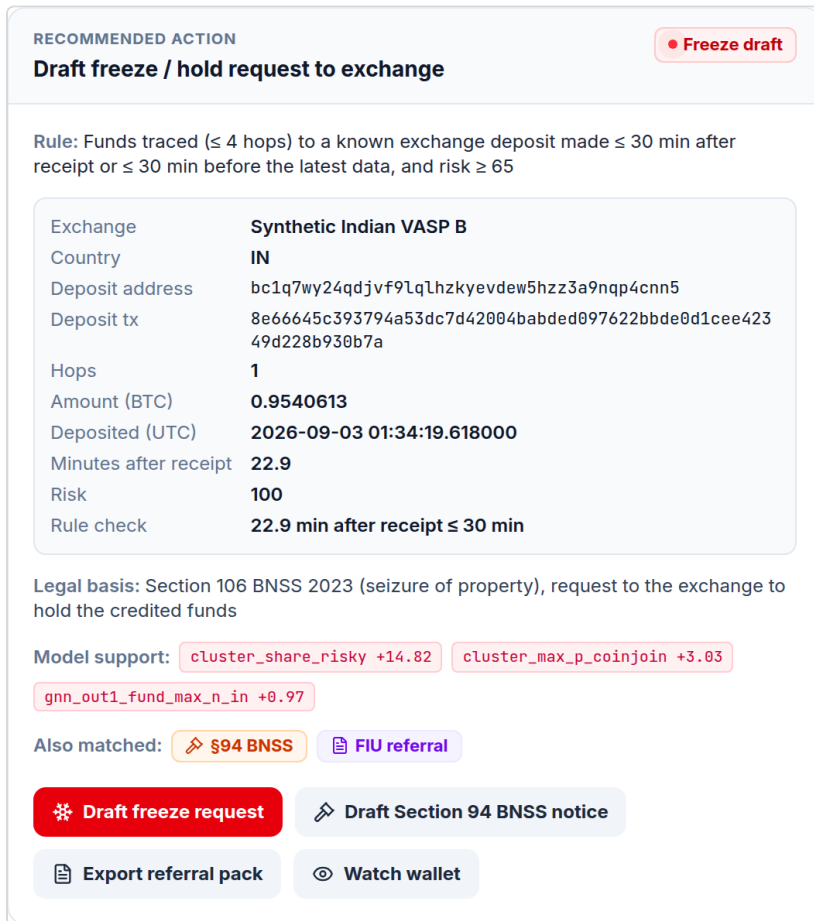


Figure 4. Action card: the directive, the rule it matched with the checked values, legal basis, model support (SHAP) and one-click drafts.



Figure 5. What would change the verdict: each factor neutralised to a typical wallet and re-scored; single-signal vs corroborated alerts.

Action directive rulebook (demo counts)

Directive	Rule	Legal basis	Alerts
Immediate Freeze Draft	Funds traced (≤ 4 hops) to a known exchange deposit made ≤ 30 min after receipt or ≤ 30 min before the latest data, and risk ≥ 65	Section 106 BNSS 2023 (seizure of property), request to the exchange to hold the credited funds	31
Draft Section 94 Bnss	Funds traced (≤ 4 hops) to an exchange that operates in India / is registered with FIU-IND	Section 94 BNSS 2023 (summons to produce documents or other things)	69
Fiu Referral Pack	≥ 1 BTC moved, broadcast via Tor / VPN / bulletproof hosting, with a layering pattern (peel chain, Coinjoin or fan-out)	PMLA 2002 intelligence sharing; the pack goes to the IO / FIU-IND liaison, not filed as an STR	0
Passive Taint Monitor	Unspent balance, dormant ≥ 6 h, linked to a seed wallet (taint > 0 or ≤ 4 hops)	No legal step yet. Watch the funds and re-alert when they move	7
Review Likely Benign	The wallet is a known exchange / mining-pool address, or is funded by a mining pool with no seed link and no risky infrastructure	None. Confirm or reject; the verdict feeds model retraining	3
Analyst Review	No directive rule matched. Read the reasons and evidence	None	26

Thresholds (freeze window 30 min, minimum risk 65, trace depth 4 hops, layering ≥ 1 BTC) live in `beans/config.py` and can be overridden without code changes. Exchange deposits come from an attribution list (`known_entities`), which is never a model feature.

Freeze / hold request (draft) A-W-35eefdf4d · bc1q4xjt2k9dkgnpc...

BEANS fills in the blockchain facts. Fields left empty stay as visible blanks for the Investigating Officer. IO details are remembered in this browser only.

Language **English** हिंदी

FIR / case no.

FIR date

Offences (sections)

IO name

IO rank

Police station

District, State

Reply within (days)

Exchange's nodal-officer address

[Redraft \(files a new request\)](#)

Evidence seal
SHA-256

PENDING SUPERVISOR APPROVAL (request #2, drafted by local). Not for issue.

REQUEST TO FREEZE / HOLD CREDITED FUNDS (SECTION 106, BHARATIYA NAGARIK SURAKSHA SANHITA, 2023)

FIR / Case No. [] dated [], P.S. [], []

To,
The Principal Officer / Nodal Officer (Law Enforcement Requests)
Synthetic Indian VASP B
[]

Subject: Immediate hold on funds credited to cryptocurrency deposit address(es): FIR / Case No. [] dated [], P.S. [], []

During investigation of the above case, registered for offences under [], proceeds suspected to be connected with the offence were traced on the Bitcoin blockchain to the following deposit address(es) of your exchange. The first deposit reached your exchange **22.9 minutes** after the flagged wallet received the funds, so the funds are likely still held by you:

- bc1q7wy24qdjvf9lq1hzkyevdew5hzz3a9nqp4cmn5
- bc1qfnt37am35093u8ac4p15tldks433qv906l87lup9jwe8nh7km4ka3nsw8v
- bc1qpsk5t43p7fvz5ryfa39vrajt93xsv8uqyp6

In exercise of the powers under Section 106 of the Bharatiya Nagarik Suraksha Sanhita, 2023, you are requested to **immediately freeze / place on hold** all balances (crypto and INR) of the account(s) to which these addresses are assigned, to block withdrawals, and to confirm the action and the balances held to the undersigned. A requisition for KYC and account records under Section 94 BNSS follows separately.

(Signature)
[], Investigating Officer
P.S. [], []
Date: []

Annex A: blockchain evidence

Flagged wallet bc1q4xjt2k9dkgnpc57f8eyy77m2mfcas9nlynmth · alert A-W-35eefdf4d ·

Figure 6. Freeze / hold request draft: blockchain facts filled in, unknown IO fields left as visible blanks, sealed with SHA-256 + RFC 3161.

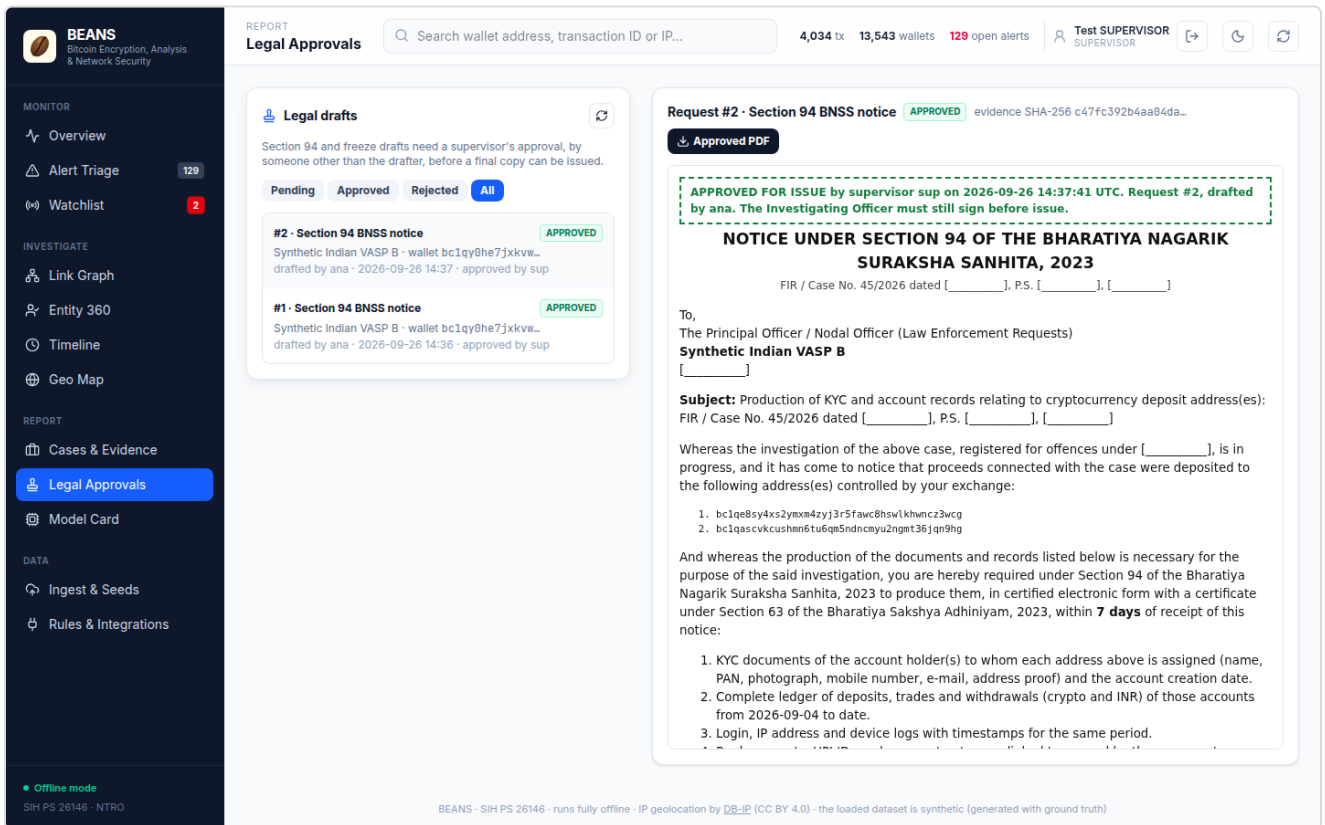


Figure 7. Legal approvals: drafts are filed pending; a supervisor other than the drafter approves or rejects (four-eyes).

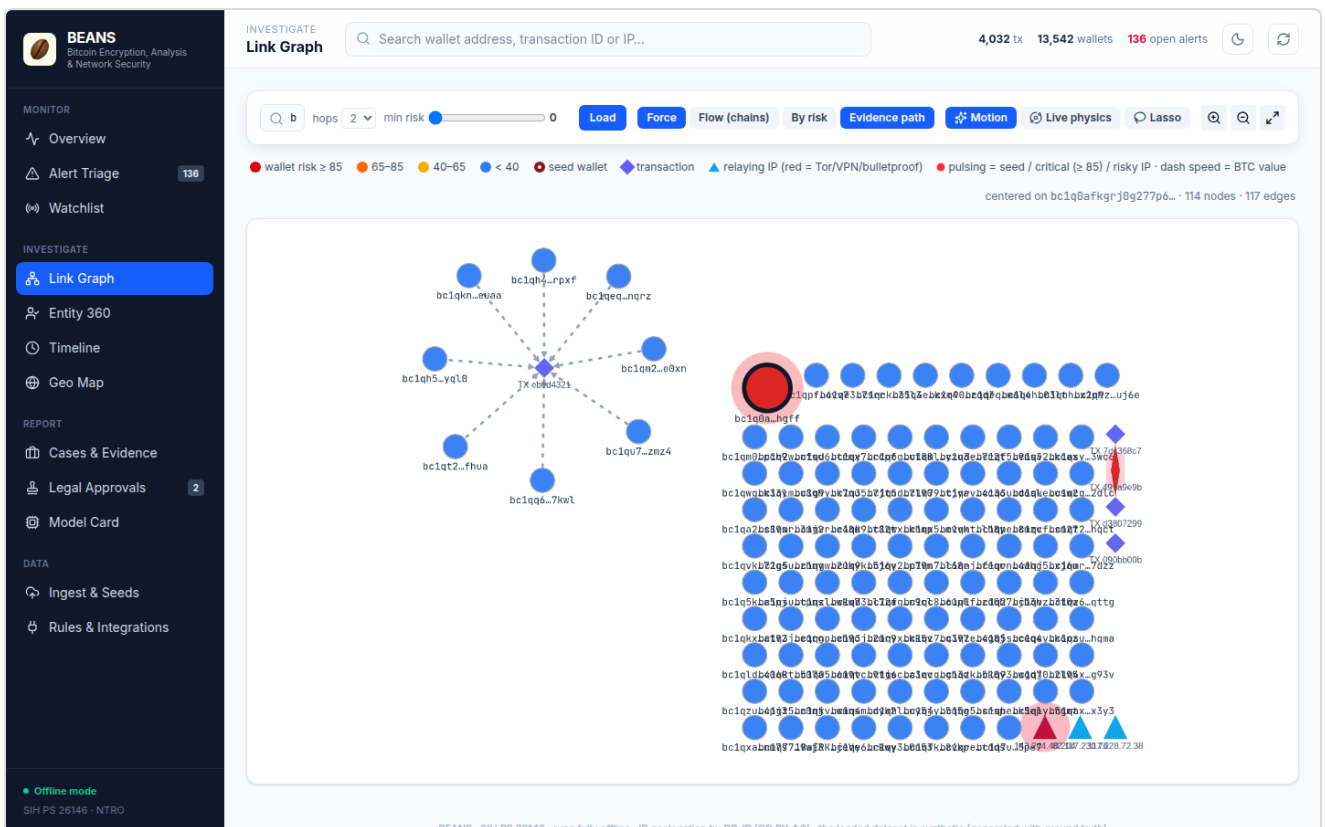


Figure 8. Link graph: money-flow particles (speed = BTC), pulsing halos on seeds / critical wallets / risky IPs, lasso selection into a case.

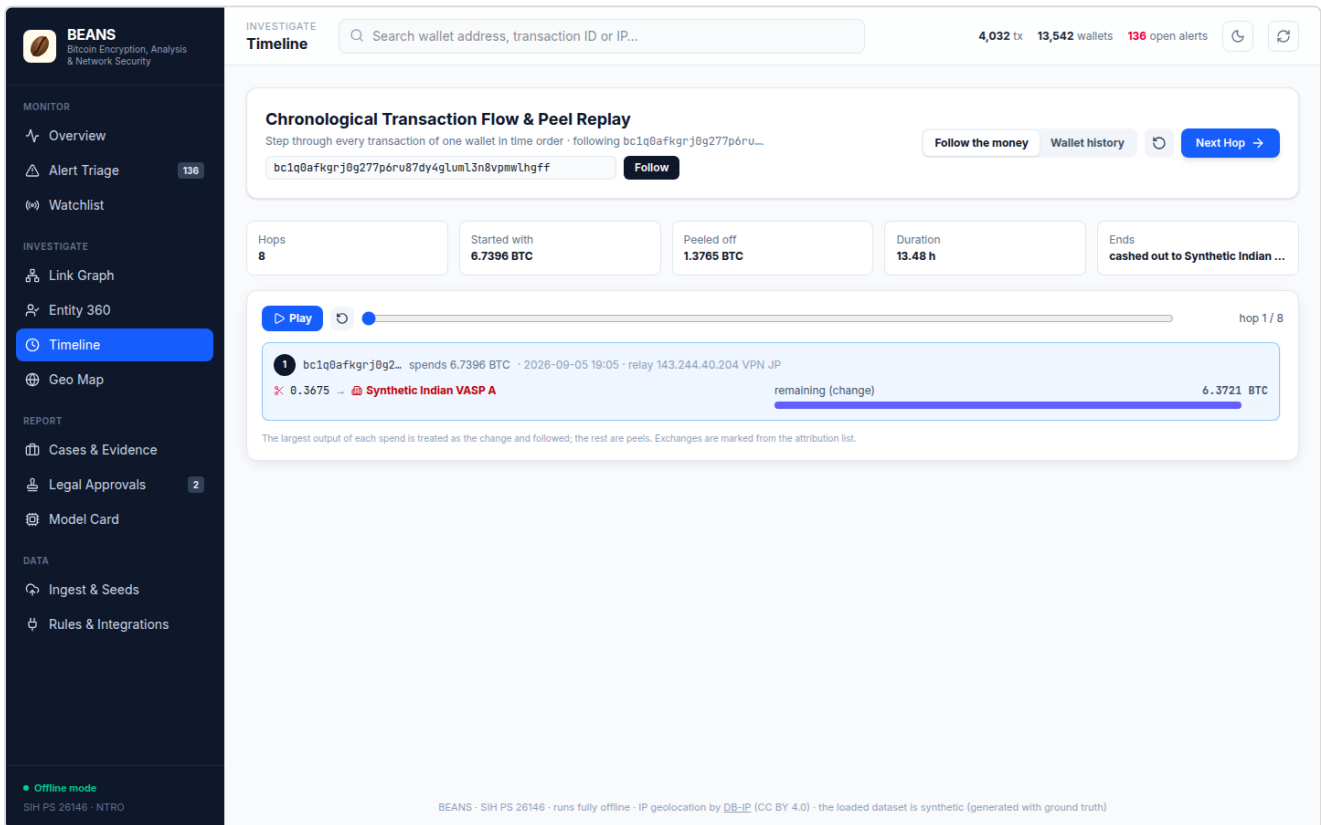


Figure 9. Follow the money: hop-by-hop peel-chain replay with peeled amounts and the exchanges they reached.

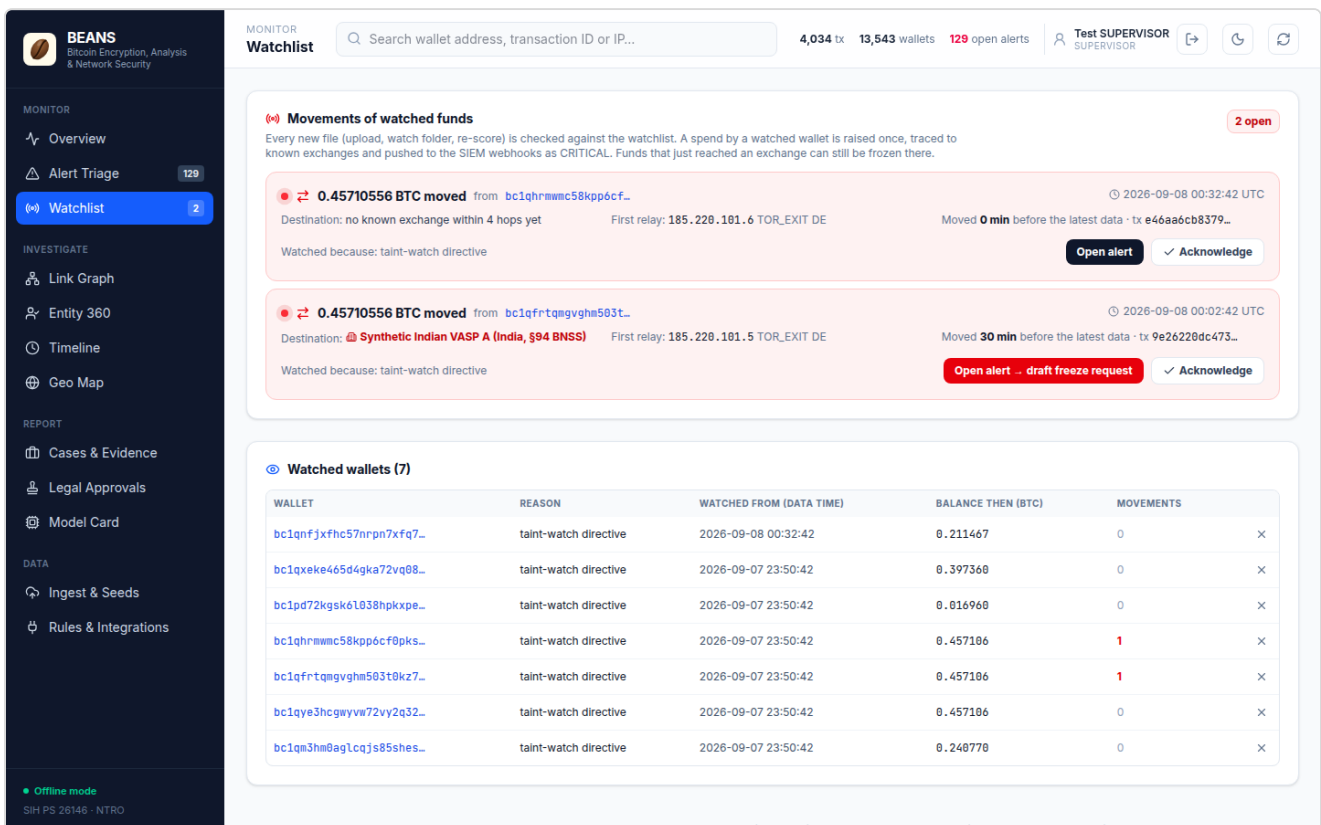


Figure 10. Watchlist: a watched wallet moved funds; one reached an Indian exchange within the freeze window.

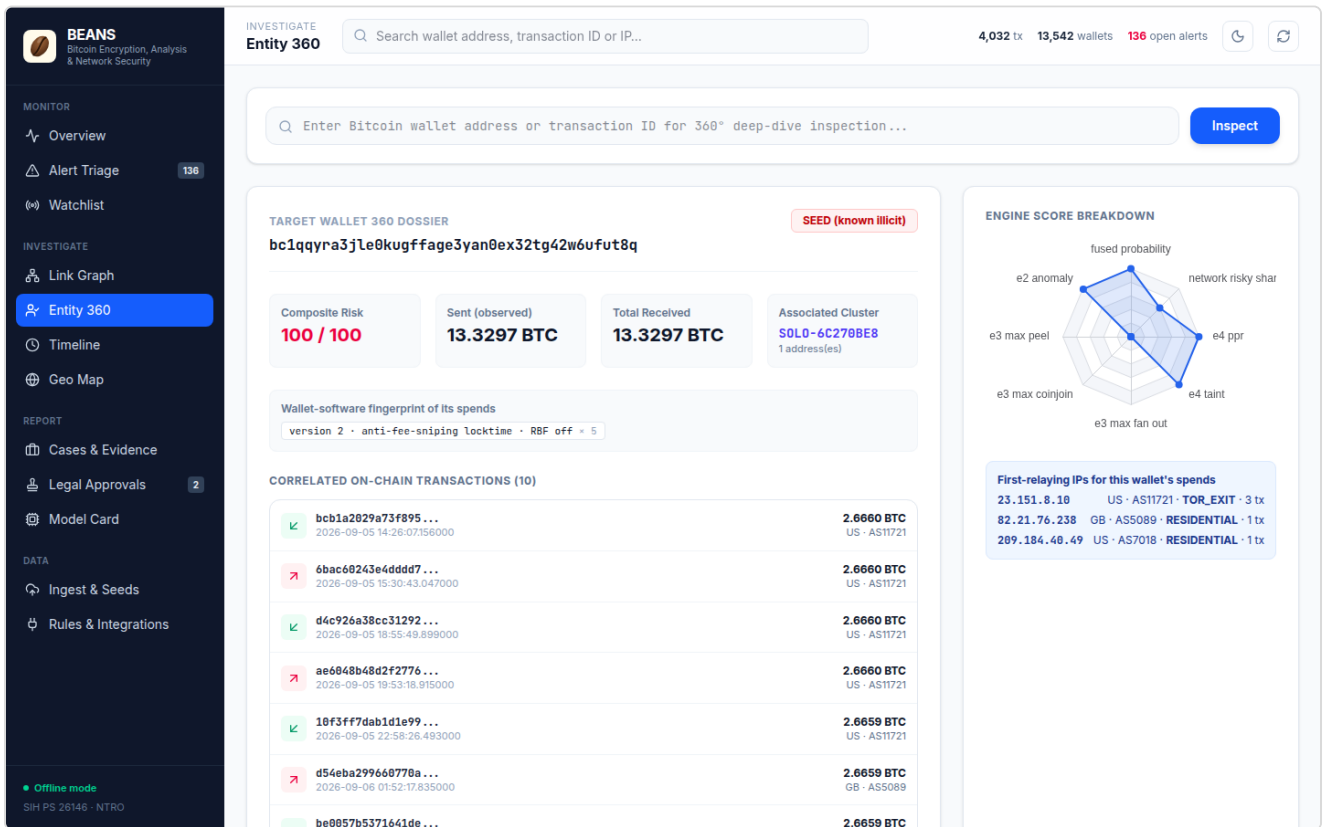


Figure 11. Entity 360: wallet profile, cluster, relaying IPs, wallet-software fingerprint and full ledger.

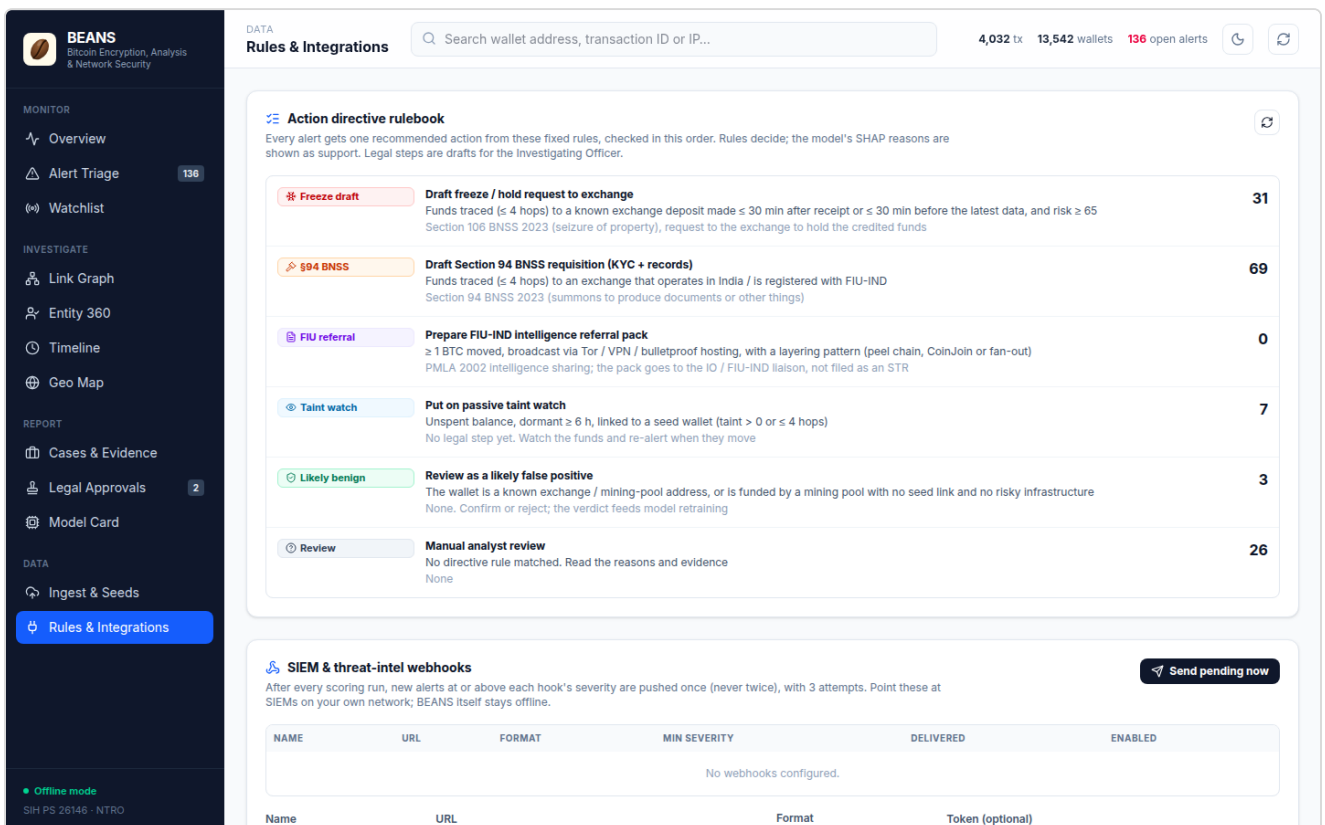


Figure 12. Rules & integrations: directive rulebook, SIEM webhooks, attribution list, evidence timestamping.

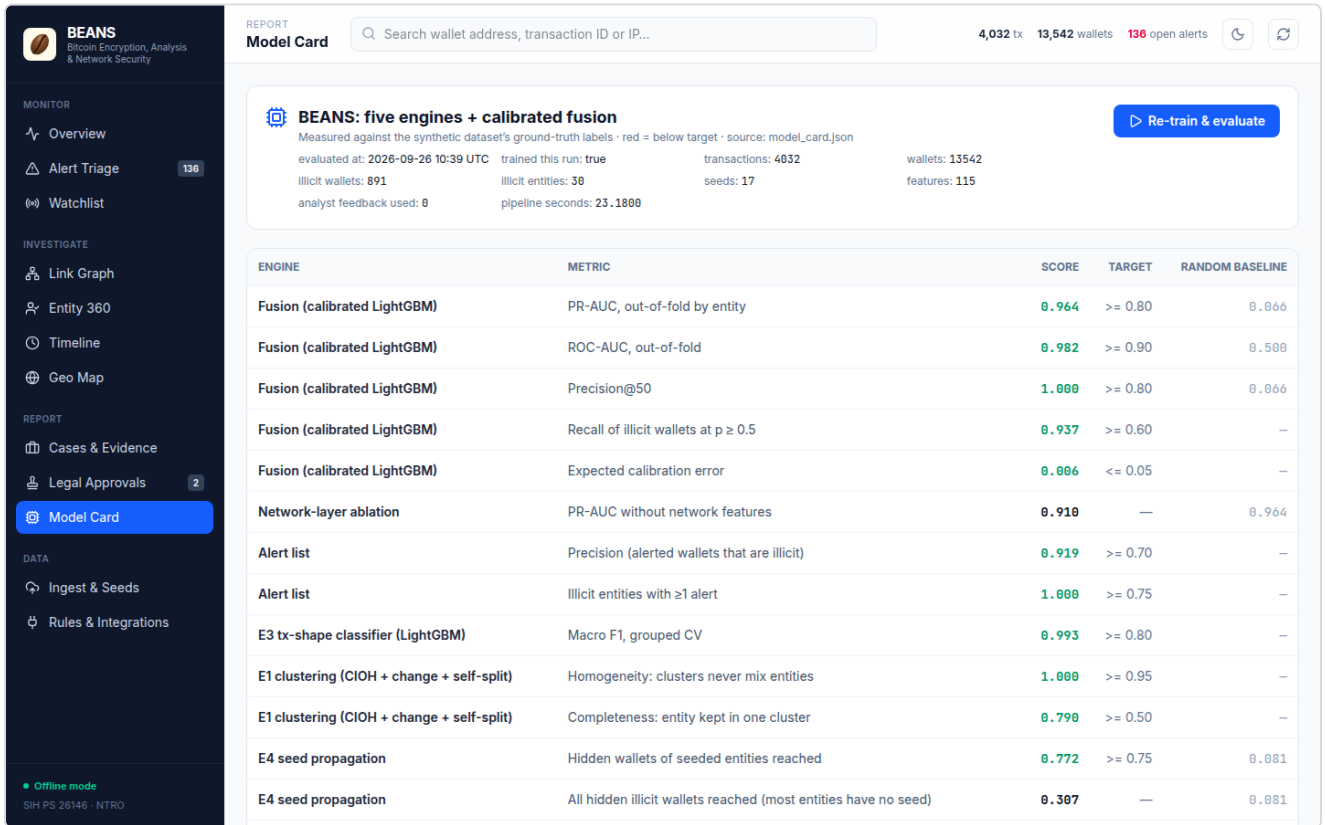
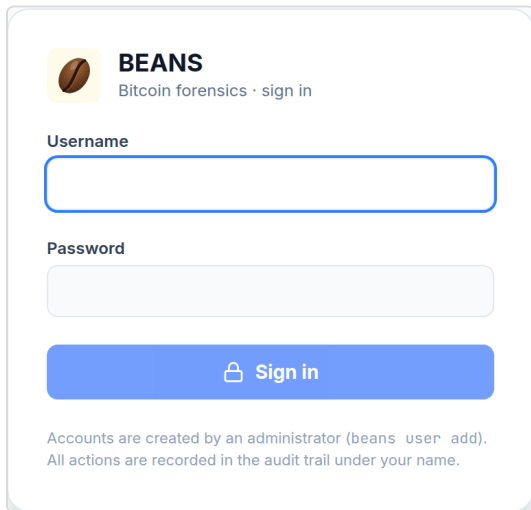


Figure 13. Model card: every metric measured on the loaded data, with targets and baselines, plus the Elliptic validation.



Figure 14. Dark theme (all pages).



The image shows a login interface for a system called BEANS. At the top left is a logo consisting of a brown circle with a white bean shape inside. To the right of the logo, the word "BEANS" is written in bold, and below it, "Bitcoin forensics · sign in" is written in a smaller font. Below the header, there are two input fields: the first is labeled "Username" and the second is labeled "Password". Both fields are empty. Below the password field is a blue button with a white lock icon and the text "Sign in". At the bottom of the interface, there is a small line of text: "Accounts are created by an administrator (beans user add). All actions are recorded in the audit trail under your name."

BEANS
Bitcoin forensics · sign in

Username

Password

 Sign in

Accounts are created by an administrator (beans user add).
All actions are recorded in the audit trail under your name.

Figure 15. Login (switches on once the first user exists).

8. Legal output and evidence

- **Section 94 BNSS requisition** (KYC, ledger, login/IP logs, linked bank/UPI, withdrawal destinations; certificate under Section 63 BSA) and **freeze / hold request** (Section 106 BNSS) in **English or Hindi**; offshore exchanges get a Letter-of-Request (Section 112) note; the freeze letter states its actual trigger (fast cash-out, or a deposit within the window before the latest data).
- **Four-eyes approval**: every draft is filed pending; only a supervisor other than the drafter can approve (or reject with a comment); the approval banner is printed on the document.
- **FIU-IND intelligence referral pack** and **case evidence packs** (PDF / HTML / JSON) with the money-trail and seed-path diagrams, counterfactuals, source-file hashes and the audit trail.
- **Sealing**: SHA-256 of the evidence plus an RFC 3161 token from a local, offline Time-Stamping Authority; anyone can re-verify with `openssl ts -verify`.

9. Monitoring, integrations and security

- **Watch folder** (`beans watch`) and **watchlist**: taint-watch wallets are watched automatically; any later spend raises one CRITICAL movement event, traced to exchanges.
- **SIEM webhooks**: generic JSON (Wazuh), Splunk HEC, Elasticsearch bulk, STIX 2.1 (MISP / OpenCTI); delivered once per alert and hook, retried, logged.
- **Users and roles**: VIEWER / ANALYST / SUPERVISOR / ADMIN; salted PBKDF2 passwords, HttpOnly sessions, lockout after repeated failures; every action in the audit trail under the user's name. No default password: single-user mode until the first user is created.
- **Cross-case links**: an alert shows when its wallet or its entity cluster is already a suspect in an open case.
- **Optional live collector** (`beans collect`): a separate tool that records transaction announcements per peer into BEANS CSVs; checked live (1,030 rows in 90 s, every txid matched the announced id).

10. Limitations

- Synthetic numbers are optimistic; Elliptic validates the modelling approach only (no addresses, amounts or IPs).
- Typology near 100 % reflects separable synthetic typologies; expect less on real cases.
- Freeze and Section 94 directives on real data depend on a real exchange attribution list.
- Legal templates need review by a lawyer; the Hindi text by a native legal reviewer (the Hindi document says so).
- The live collector can only resolve inputs from transactions it has already seen (12 of 311 after 90 s); a full node's export gives complete inputs.
- The local TSA proves "no later than" by this machine's clock and key; it carries less weight than an accredited public TSA.

Appendix: reproduce

What	Command
Demo (offline)	<code>make install && make demo</code> → http://127.0.0.1:8000
Offline Docker test	<code>make docker-offline-test</code>
6-dataset benchmark	<code>.venv/bin/python scripts/evaluate_seeds.py --seeds 42 7 123 2024 99 555</code>
Real-data validation	<code>beans validate-elliptic --download --doc docs/VALIDATION_ELLIPTIC.md</code>
1M-row benchmark	<code>.venv/bin/python scripts/benchmark_1m.py --files 10</code>
This report	<code>.venv/bin/python scripts/build_report.py</code>
Tests	<code>make test</code> (62 tests)

IP geolocation by DB-IP (CC BY 4.0). Elliptic dataset: Weber et al., "Anti-Money Laundering in Bitcoin", KDD 2019 workshop (arXiv:1908.02591). SIGN: Frasca et al. 2020. All data shown in screenshots is synthetic.